

HelixLLM Phase-3 CPU

Translation (NMT) — END-TO-END PROOF (§11.4.108)

Status	COMPLETE — ALL-GREEN. Real CPU translation service booted via the containers submodule, anti-gaming triple + determinism + analyzer self-validation all PASS, torn down single-owner, coder untouched.
Run-id	phase3_translation_20260707
Date	2026-07-07T07:05:03Z → 07:05:59Z (UTC), host x86_64, podman 5.7.1, rootless
Branch / Track	feature/helixllm-full-extension · (T1/main)
Design	docs/research/ 07.2026/00_master/ TRANSLATION_PROVIDER.md (commit c9ac8683) §4 acceptance signature
Pattern mirrored	docs/qa/ phase3_embeddings_20260706/ harness/ (embeddings proof, commit e46df52c)
Lane	stock LibreTranslate (Argos/ OPUS-MT on CTranslate2, CPU) — the design's DOCUMENTED FALLBACK lane. Honest §11.4.6 substitution recorded in 23_substitution.txt + §“Lane decision” below.

1. Lane decision — honest substitution (§11.4.6)

- **Design PRIMARY (deferred):** NLLB-200-distilled-600M CT2 int8 behind a bespoke LibreTranslate-shaped /translate FastAPI shim (TRANSLATION_PROVIDER.md §1.2). That lane needs a custom image build (python + ctranslate2 + sentencepiece + FastAPI + a language detector) **plus** an offline HF → CT2-int8 model conversion — a substantial artefact deferred as its own work item.
 - **Lane PROVEN here:** stock **LibreTranslate** — the design's own §1.2/§3.1 **documented fallback lane:** the *identical* LibreTranslate / translate contract on the *same* CTranslate2 engine the primary would use, delivered as a single pre-built CPU image (615 MB), so it boots reliably with zero new shim code. The task explicitly sanctions this as “the simplest reliable CPU lane.”
 - **This is a REAL NMT service** (real Argos/CTranslate2 translation). The anti-gaming triple below proves genuine translation — **not** an identity / passthrough bluff. A faked/identity-copy PASS is impossible here (proven by the RED baseline and the golden-bad self-validation).
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2. What was booted, and how (§11.4.76 / §11.4.161, NO GPU)

- Booted **through the containers-submodule compose.Orchestrator** (digital.vasic.containers/pkg/compose) — NOT ad-hoc podman/docker. Rootless podman. Compose file harness/compose.phase3translate.yml carries **no literal** host/port/model/language value (§CONST-045/046); every value is env-injected by run_proof.sh.
- **NO GPU** in the run spec (no --device nvidia.com/gpu, no GPU flag) — the structural guarantee this ships before the P0/P1 GPU chain. compute_type stays CPU (Argos/CT2 CPU kernels).
- Image docker.io/libretranslate/libretranslate:latest (615 MB), host port **18436** (distinct from coder :18434, embeddings :18435), container port 5000.
- LT_LOAD_ONLY=en,fr,de → /languages served exactly [en, fr, de] (25_languages.json), so cold boot fetched only the en/fr/de Argos packages.

- Persistent external Argos-model cache volume helixllm-libretranslate-cache (§11.4.77 re-obtainable; survives teardown so re-runs skip the download).
- **Boot lessons reused from the embeddings proof:** no WithForceRecreate (leaves the pod unstarted on this host's podman-compose shim); unique per-run project name + pre-clean boot-down for a genuinely fresh clean-target; health poll fast-fails **only** on container state=exited (created/missing keep polling); persistent external model-cache volume. Boot became healthy (/languages → 200) after **11 polls** (21_health.txt).

3. §11.4.108 RUNTIME SIGNATURE — the CX-05 anti-gaming triple (GREEN)

For each pair, the harness drove the REAL service: forward POST /translate, POST /detect on the forward text, then a back POST /translate, and asserted **all three** criteria (no single one trusted — the CX-05 anti-gaming construction):

1. **Not-identity + detected-target-language** — forward $\neq$ source AND /detect(forward) == target.
2. **Forward adequacy** — chrF(forward, independent golden reference) ≥ 0.30 .
3. **Back-translation metamorphic** — chrF(back, source) ≥ 0.40 .

chrF = sacrebleu-style character-n-gram F-score (char_order=6, $\beta=2$, whitespace-stripped); deterministic, reference-based, language-agnostic. Thresholds calibrated on this project's own fixtures (§11.4.107(13)) — real output clears them with wide margin, every golden-bad falls far below.

Captured results (real /translate + /detect responses)

Pair	source	forward (real Argos output)	detected	back	fwd chrF (floor 0.30)	back chrF (margin 0.40)	verdict
en → fr	"The book is on the table."	"Le livre est sur la table."	fr (conf 100)	"The book is on	1.0000	1.0000	PASS

Pair	source	forward (real Argos output)	detected	back	fwd chrF (floor 0.30)	back chrF (margin 0.40)	verdict
				the table.”			
en → de	“The book is on the table.”	“Das Buch liegt auf dem Tisch.”	de (conf 100)	“The book is on the table.”	0.7519	1.0000	PASS

notIdentity=true, targetMatch=true for both. Evidence:

11_green_proof_en_fr.txt, 11_green_proof_en_de.txt, raw records
green_record_en_{fr,de}_1.json.

The en → de forward “Das Buch **liegt** auf dem Tisch.” (“lies”) differs from the golden “Das Buch **ist** auf dem Tisch.” (“is”) — both are correct German; chrF 0.7519 correctly scores it well above floor. This is genuine calibrated adequacy, not a tuned-to-match golden.

4. Determinism (§11.4.50) — GREEN

Two byte-for-byte identical requests per pair produced **byte-identical**

forward + back text: - [DETERMINISM] PASS ... (en->fr) —

11_green_proof_en_fr.txt - [DETERMINISM] PASS ... (en->de) —

11_green_proof_en_de.txt

5. RED baseline (§11.4.115) — defect reproduced, analyzer is not a bluff gate

10_red_baseline.txt: the identity-passthrough “warming” bluff (gateway echoes q untranslated — the exact defect design §2.5 forbids) was fed to the analyzer with RED_MODE=1 and correctly **FAILED** (exit 1):

```
[RUNTIME-SIGNATURE(en->fr)] FAIL notIdentity=false
```

```
targetMatch=false fwdChrF=0.2242(floor 0.30) backChrF=1.0000
```

```
reason: identity/passthrough: forward "The book is on the  
table." == source "..."
```

```
reason: detected language "en" != requested target "fr"
reason: forward chrF 0.2242 < floor 0.3000
```

Note: back-translation alone (chrF 1.0000) would PASS the identity copy — criteria 1 and 2 are what defeat it. This is the CX-05 anti-gaming guarantee in action.

6. Golden-good / golden-bad analyzer self-validation (§11.4.107(10)) — GREEN

12_self_validation.txt — golden-good = the REAL captured en → fr record; every golden-bad variant MUST FAIL (proves the analyzer itself cannot be fooled):

Fixture	expect	fwd chrF	back chrF	result
GOLDEN-GOOD (real en → fr)	PASS	1.0000	1.0000	PASS ✓
BAD identity/passthrough	FAIL	0.2242	1.0000	FAIL ✓ (crit 1+2)
BAD wrong-language (de for fr)	FAIL	0.0873	0.1859	FAIL ✓ (crit 1+2+3)
BAD garbage (with faked target-match)	FAIL	0.0500	0.0269	FAIL ✓ (crit 2+3 catch it despite faked crit 1)
BAD empty	FAIL	0.0000	0.0000	FAIL ✓ (all)

The garbage fixture deliberately **fakes** targetMatch=true to prove the forward- chrF + back-metamorphic criteria independently catch it — no single criterion is trusted (the §4.1 paired-mutation guarantee). Floor 0.30 cleanly separates real (0.75–1.00) from every bad fixture (0.00–0.22).

7. Single-owner teardown + coder untouched (§11.4.119)

29_teardown.txt / 29b_post_teardown.txt: - DOWN-OK via the containers-submodule orchestrator; phase3translate_lt containers removed ((none – removed)). - **helixllm-coder** — Up 17 hours before, during, and after — never touched. - A concurrent, unrelated helixllm workstream container (llamacpp-router VLM model-download) was observed on the host and left untouched (§11.4.174 — ownership verified before any action; never mine to stop).

8. Four-layer verification map (§11.4.108)

1. **SOURCE** — harness/
{main.go,compose.phase3translate.yml,run_proof.sh} committed (route/contract driven).
2. **ARTIFACT** — 01_image_pull.txt (615 MB image pulled by digest tag); /languages = [en,fr,de].
3. **RUNTIME-ON-CLEAN-TARGET** — pre-clean boot-down + unique project name ⇒ fresh container; the §3 triple verified against it.
This is the definition of done.
4. **USER-VISIBLE** — a real client POST to /translate returns a correct, right-language translation (Le livre est sur la table. / Das Buch liegt auf dem Tisch.); /detect confirms the target language; the round-trip recovers the source.

9. Honest boundary (§11.4.6)

- This proves a **real CPU translation service** passing the anti-gaming triple + self-validation for **two** pairs (en → fr, en → de). It does **not** prove the design PRIMARY (NLLB-200-CT2 shim) — that lane is deferred (see §1).
- chrF is a character-n-gram adequacy proxy vs a golden reference; it is the deterministic automated floor the design §4 mandates (COMET/QE is a later additive signal, never a replacement).
- Two independent human golden references were authored **before** observing model output; en → fr happened to match exactly (chrF 1.0), en → de scored 0.75 against a correct-but-differently-phrased reference — i.e. the floor is a genuine adequacy separator, not a tuned-to-model artefact.

10. Reproduce

```
cd docs/qa/phase3_translation_20260707/harness
./run_proof.sh          # builds harness, pulls image, boots via
                        containers
                        # submodule, proves, tears down single-
                        owner. Regenerates
                        # every evidence file in the parent dir.
```

Config knobs (env, all default-injected): LT_HOST_PORT (18436),
LT_LOAD_ONLY (en,fr,de), CHRF_FLOOR (0.30), BACK_MARGIN (0.40),
LT_MEM_LIMIT, LT_CPUS.

11. Evidence index

File	What
00_preflight.txt	host, coder-untouched, free port, config
01_image_pull.txt	image pull (ARTIFACT layer)
10_red_baseline.txt	§11.4.115 RED — identity passthrough FAILs
11_green_proof_en_fr.txt / _en_de.txt	§11.4.108 triple + determinism per pair
12_self_validation.txt	§11.4.107(10) golden-good/bad analyzer self-validation
13_verdict.txt	ALL-GREEN
20_boot.txt / 21_health.txt	boot via orchestrator + health OK (11 polls)
23_substitution.txt	honest lane substitution note (§11.4.6)
24_container_state.txt / 25_languages.json	running container + served languages
29_tearardown.txt / 29b_post_tearardown.txt	single-owner teardown + coder untouched
30_probe_en_{fr,de}.txt	raw probe transcripts (source → forward → detect → back)
green_record_en_{fr,de} _{1,2}.json	raw /translate+/detect responses (determinism pair)
red_identity_record.json	the RED identity-passthrough fixture

File	What
harness/	main.go (analyzer+client), compose.phase3translate.yml, run_proof.sh, go.mod, .gitignore